Algonox

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REVIEW PRESENTATION

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1.1 Brief Introduction :

- My name is **Anil Kumar**, and I recently completed my **B.Tech in Computer Science and Engineering** branch in **IIIT Srikakulam** in 2025. I joined **Algonox Technologies** in **December 2024** as a **Python Backend Developer Intern**.

- During my internship , I have been actively involved in **Building, Enhancing**, **Debugging** and **Resolving** Backend APIs & integrating backend functionalities with multiple **Databases** to ensure smooth data flow across all the **Tenants** and writing clean , efficient code with all the Application coding standards.

1.2 Assigned Projects and Responsibilities

- **Project** : AMOP 2.O

- **Responsibilities** :

- **Building & Enhancing** Backend APIs (Bulk Change, Inventory, ReadMe, Carrier Syncs, Reports , and Dashboard modules.)
- **Debugging & Fixing** Issues(Prod/Beta-Level) : QA check list bugs.
- **Readme** API Documentation & Postman Collections.
- Taking **Ownership & Accountability** of my Tasks.
- Knowledge Transfer (**KT**) to New Interns: local codebase setup + Inventory + Bulk Change UI + Readme workflows.

2.1 Challenges Faced and How I Addressed Them

1. **Dashboard Data Delay** (performance optimization) **-** Redis Cache.
2. **Bulk Change Handling -** Thread Pool Executor + Batch Processing.
3. **All Change Types Upload – Scalability Enhancement -** Multi Threading.
4. **Future Effective Date Handling -** Amazon Event Bridge.
5. **Carrier sync all devices processing parallelly -** Retry attempts + ThreadPool Executor + Batch Processing.

2.2 Impact of My Work

1. Readme APIs and related postman collection for Bulk change + Inventory APIs : **4488**
2. Pond IoT (4) change types : **3820 ,3821 , 3822 , 3823**
3. Release of Demo customer Tenant(**API Test Tenant**).
4. All Files cleaning with all Application standards : Logs, Auditing ,sql injection prevention , Doc strings , comments.
5. Implemented Multi RDS for all the modules.
6. Dashboard charts : **Sim vs Service Provider & Service Provider vs Activation Trend & Service Provider vs Status**
7. User timeout setting in Super Admin module **=>**Partner Module : **4432**
8. Auditing Tracking of APIs across application - **Excel file prepared**.
9. Update device status bulk upload in the bulk change module -
10. Bulk upload all change types in Bulk Change module - **5002**
11. History tables updation - **4940**
12. Main tenant customer group creation in new tenant creation by uploading and an excel file - **5053**
13. Main Wrapper Function to process new tenant creation Excel template(Main tenant) - **4925**

2.2 Impact of My Work

14. Created active lines report in Reports module - **4211**
15. Revenue Assurance - Add a Username Column - **4767**
16. Fixing QA check list Bugs -
17. Implemented Assign Customer change type for all service providers in Inventory - **4594**
18. Line sync change type for all service providers in inventory module - **4598**
19. PPU address change type in inventory module - **4589**
20. Refresh a line change type in inventory module - **4599**
21. Implemented communication plan change type for all jasper providers in **Bulk change UI + Upload** – **5136**
22. Implemented communication plan change type for all jasper providers in **Inventory Module** - **5137**
23. Future Effective Date handling – **5261**
24. Implemented Carrier Sync Processor - **5187**
25. Kore Carrier Sync – **5181, 5208, 5210**
26. Carrier sync Email Trigger At sync Initiation , Completion , Error cases – **5212**
27. Amop 2.0 API - Usage By Line Report – Timeout **- 5294**
28. Sub Tenant(wizard) Data Insertion Based on Groups,Service Providers - **5356**

3.1.A) Technical Skills :

1. **Programming :** Python, SQL, Flask.
2. **Deployment :** AWS Lambda , EC2.
3. **Database Handling :** Postgres SQL.
4. **Debugging :** Cloud watch logs and Putty.
5. **Tools :** Postman, Jira, Gitlab, Github, Github Copilot, Vs code.

3.1.B) Soft Skills :

1. **Explaining my work** in Sync calls.
2. **Cross-functional collaboration :** with Dev , UI , QA and BA teams.
3. **Adaptability:** Quickly learning and working on diff. modules and deployment platforms Lambda & EC2 , different tasks implementation / debugging / resolving Issues.
4. **Time Management:** I will use time productively and by Balancing implementation , debugging & fixing production/UAT level issues parallelly(on same day) on priority basis.
5. **Following App. Standards** : By strictly adhere document workflow and all coding standards like auditing,logging,comments and writing testcases.
6. **Ownership & Accountability** of my work**.**

3.2) How I Applied Feedback to Improve :

1. **Code review Feedback** : Write clean, readable, understandable ,efficient code with reusable functions , exception handling , auditing, doc-strings , structured logs , Reduced hardcoding , prevention of SQL injection vulnerabilities.
2. **QA Testing Feedback :** Increase test case coverage, especially for edge cases & to reduced no.of back-and-forths b/w Dev and QA teams.
3. **Lead/seniors Feedback :** to understand complete Requirements & Design workflow before new implementation & Strictly adhered to design doc. by following all the coding standards & utilize time more productively by focusing on priority tasks first.

4.1) Career Goals(Next 1–2 Years) :

1. **Evolving into a Senior Backend Developer :** With strong backend knowledge & solving complex backend problems independently, writing high-quality code , taking more technical responsibilities.
2. **Contributing to Enterprise-Grade Systems** : Focus on Building scalable, secure, and high-performance backend systems that can handle large amounts of real-world data and user requests reliably.
3. **End-to-End Project Ownership :** From requirements understanding to development, testing, deployment and production monitoring by Ensuring that the solution is reliable, scalable, and aligned with all project standards while delivering high-quality outcomes on time.
4. **Continuous Learning & Innovation :** Stay updated with modern backend technologies, cloud platforms, and scalable architecture patterns to continuously bring innovative and efficient solutions to my organization.
5. **Leadership & Mentorship :** want to guide junior developers and interns through structured knowledge transfers.

4.3) Contributions to the Organization on Fulltime Role:

1. **Maintain Applications Stability & Standards** : By Ensuring all application strictly adhere to documented workflows and all the coding standards.
2. **Taking Ownership and Accountability :** assigned tasks/modules from implementation to deployment and issue resolution , Ensure tasks are delivered on time with high quality.
3. **Contributing to Long-Term Product Growth & Reliability :** Focusing on building solutions that handle large volumes of data, improve system performance, and support future feature expansion than just fixing issues immediate without having long-term vision.
4. **Cross-Team Collaboration :** Collaborate with Development, QA, and BA teams to ensure smooth communication, faster issue resolution.
5. **Supporting and Guiding New Interns :** with local codebase setup, Application workflow understanding ,to help them quickly to adapt to the application codebase & actively share my knowledge with my team.

4.3) Summary of My Internship Experience :

- During my internship at **Algonox Technologies**, I worked on the **AMOP 2.0** Project using Python, SQL , PostgreSQL and contributed to backend development across modules like Bulk Change, Inventory, Carrier Syncs, Dashboard, Reports and Partner Addition. I was responsible for **developing, enhancing, debugging, and resolving** backend APIs in a real production environment. I also implemented **performance optimizations and scalable solutions** such as **Redis caching, multithreading, Thread pooling , batch processing, and event-driven scheduling.**

- Along with feature development, I followed enterprise development practices like **code reviews, QA/UAT testings , Auditing, Design document workflow , and secure coding practices**. Additionally, I gained exposure to cloud platforms like **AWS Lambda and EC2,** handled production level debugging using CloudWatch and Putty, and collaborated with **Development, QA, and BA teams** to ensure smooth communication, faster issue resolution, and better alignment between requirements and implementation.

Thank You
for Your Time and Attention

— Kumar Anil